# UNIT 15 CENTRAL BANKS AND THE SUPPLY OF MONEY*

**Structure**



## 15.0 OBJECTIVES

After going through this Unit you should be in a position to

- discuss how the supply of money is measured;
- identify the components of money supply;
- explain how changes in money supply take place; and
- analyse the scope of open market operations.

## 15.1 INTRODUCTION

Many years ago, around 6000 BC, when money was not invented, there was the barter system. Under this system, people used to trade goods and services with one another in the absence of money. As money was not invented, people exchanged goods and services for other goods and services in return. The disadvantage of this system was that one could exchange items only when there was a 'coincidence of wants' (also called double coincidence of wants). For example, I have wheat with me and I want apple in exchange. I should look for a person who has apple and wants to exchange it for wheat. This sort of coincidence, as you could imagine, was rare and created difficulties in transaction of goods and services. When money got invented, exchange process became smooth enough.

Money is anything that is commonly accepted by people for the exchange of goods and services. Every country has its own system of currencies in the form of coins and paper notes. Money in its present form however is a recent phenomenon. In the olden days, there were other entities which were considered as money. There was a lot of experimentation in the search for an ideal commodity which could serve the purpose of money. In this experimentation


* Dr. Jagannath Mallick, Independent Researcher, Delhi

there were animals, valuable commodities and precious metals. Salt, seeds, cattle, tea, etc. were popular commodities to be used as money in the past. Commodity money provided intrinsic value, but it faced certain difficulties in terms of transportation, storage, and lack of longevity. This led to introduction of coins made of precious metals such as gold and silver as money. Finally, non-precious materials have been used to make money, as we see today (such as coins and paper currencies). These are known as fiat money – they have neither intrinsic value nor can they be redeemed for any precious commodity. A currency note can be exchanged only with currency notes of other denominations.

There are certain criteria that money should fulfil, viz., (i) it should be amenable to standardisation so that all units are uniform, (ii) it should be acceptable by all, (iii) it should be divisible so that it can be exchanged with currencies of other denominations, (iv) it is easy to carry, and (v) it has certain longevity. Commodity money did not fulfil most of these criteria while the present day paper money fulfils all the criteria.

Money has three important functions to play, viz., (i) unit of account, (ii) medium of exchange, and (iii) store of value. Price of a commodity, which represents its market value, is expressed in terms of money. Thus money acts as a unit of account or a measure of value. Further, money can be used any time, at present or in future, to purchase any commodity. Thus it can be stored, from the time it is received till it is spent on purchase of some goods and services. Thus, having money entitles a person to certain purchasing power. Since each and every commodity is expressed in terms of price, it is easier to exchange goods and services. In fact, every transaction, whether selling or buying, can be seen as an exchange between the commodity concerned and money. Thus, money eliminates the problem of coincidence of wants and the process of exchange of commodities becomes smooth.

## 15.2 MEASURES OF MONEY SUPPLY

Money is issued by the Central Bank of the country concerned. In India, for example, the Reserve Bank of India (RBI) issues currency notes of several denominations. Coins of several denominations are issued by the Government of India, but circulated by the RBI. The RBI is responsible for design, production and management of currencies in the country. By money supply we mean the total amount of currency and other liquid instruments circulating in the country.

You should note that there is no unique definition of 'money supply', either as a concept in economic theory or as measured in practice. Money is defined by the functions it performs. Thus when we look into its 'store of value' function, there could be several instruments of storing money – as cash in hand, as savings deposit in a bank, as fixed deposit in a bank or post office, etc. Remember that each of these methods have different levels of liquidity. Similarly, acceptability of money as a 'medium of exchange' is not limited to currency in circulation. You can make payments through several methods such as cash, cheque and electronic transactions – each one being different in terms of liquidity. Remember that cheque and electronic transactions are backed by fiat money.

Since these instruments perform the functions of money, it is not appropriate to restrict money supply to currencies in circulation only.

Total stock of money includes all the money held with the public, the RBI as well as the government. By money supply we mean the first component only (that is, money held with the public). Here, the word 'public' includes the households, firms, local authorities, and companies. Money held by the RBI and the government are not included in money supply. Thus money supply does not include money held as cash reserve ratio (CRR) with the RBI, statutory liquidity ratio (SLR) with the commercial banks, and money held by the government as these are out of circulation.

Let us look into the definition of various measures of money stock in India. Since July 1935, the RBI is the sole authority to compile and disseminate the statistics on monetary indicators in India. Based on the changes in the structure of the Indian economy various working groups were set up over the years to review and redefine the monetary aggregates. The First Working Group on Money Supply (FWG) was set up in (1961), the Second Working Group (SWG) was formed in (1977) and the "Working Group on Money Supply: Analytics and Methodology of Compilation" (WGMS) was formed in (1998). The Monetary aggregates are published on a regular basis in various publications of the RBI such as Annual Report, Handbook of Statistics, and the RBI Bulletin.

From policy perspectives, money can be defined as the set of liquid financial assets, whose variation in the stock could influence aggregate economic activity. The quantification and study of the money supply directs the policy makers and economists to design the policy of changing money supply according to the situations in an economy. Thus, a range of monetary and liquidity measures are compiled in India (see Table 15.1 for their definitions).

**Table 15.1: Measures of Monetary and Liquidity Aggregates**

| | | |
|---|---|---|
| M0 | = | Reserve Money |
| | = | Currency in circulation + Bankers' deposits with RBI +'Other deposits' with the RBI |
| | = | Net RBI credit to the Government + RBI credit to the commercial sector +RBI's claims on banks + RBI's net foreign assets + Government's currency liabilities to the public – RBI's net non-monetary liabilities |
| M1 | = | Currency with the public + Demand deposits with the banking system + 'Other deposits' with RBI. |

| | | |
|---|---|---|
| M2 | = | M1 + Savings deposits of post office savings banks |
| M3 | = | M1+ Time deposits with the banking system |
| M4 | = | M3 + All deposits with post office savings banks (excluding National Savings Certificates) |
| NM0 | = | Monetary base |
| | = | currency in circulation + Bankers deposits with RBI + 'other deposits' with RBI |
| NM1 | = | Currency with the public + Demand deposits with the banking system + 'Other deposits' with the RBI |
| NM2 | = | NM1 + Short-term time deposits of residents (including and up to the contractual maturity of one year) |
| NM3 | = | NM2 + Long-term time deposits of residents + Call/ Term funding from financial institutions |
| L1 | = | NM3 + All deposits with the post office savings banks (excluding National Savings Certificates) |
| L2 | = | L1 + Term deposits with term lending institutions and refinancing institutions (FIs) + Term borrowing by FIs + Certificates of deposit issued by FIs |
| L3 | = | L2 +Public deposits of the non-banking financial companies |

**Notes:** *Net Bank Credit to the Government* = Net RBI Credit to the Government (*i.e.,* Net RBI Credit to the Centre + Net RBI Credit to State Governments) + other banks' credit to the Government

*Bank Credit to the Commercial Sector*= RBI credit to the commercial sector + other banks' credit to the commercial sector

*Net foreign assets of the banking sector* = RBI's net foreign assets + other banks' foreign assets

*Net non- monetary liabilities of the banking sector* = RBI's net non-monetary liabilities + Net non-monetary liabilities of other banks.

***Source: RBI***

You should note that M0 is known as 'monetary base', 'reserve money', 'central bank money', or 'high-powered money'. The monetary measure M1 includes M0, demand deposits with commercial banks, and other deposits with the RBI (such as deposits of other financial institutions apart from banks, internal amount pending adjustments, etc.). The measures M0 and M1 are sometimes called 'narrow money' as they indicate money supply that can be easily converted into cash. If we add post office saving bank deposits to M1, we obtain M2. Further, if we add time deposits (also called fixed deposits) to M1 we obtain M3. If we add total post office deposits to M3, we obtain M4.

While M0, M1 and M3 are widely used in India, M2 and M4 are rarely used. The RBI started publication of new set of monetary aggregate (NM0 to NM3) and liquidity aggregates (L1 to L3) following the suggestions of WGMS in 1998. The measures NM2 and NM3 are applicable to the residents of the country. Currency and deposits held by non-resident Indians are normally recorded in the balance of payments accounts as international capital flows. However, it may not be appropriate to exclude all categories of non-resident deposits from domestic monetary aggregates as non-resident rupee deposits are essentially integrated into the domestic monetary market. Hence, the non-resident repatriable foreign currency fixed deposits such as foreign currency non-resident (bank) or FCNR (B), Resurgent India Bonds (RIBs), and India Millennium Deposits (IMDs) are excluded from deposit liabilities and they are treated as external liabilities. Other features of the new monetary and liquidity aggregates are as follows:

1. There are the four measures of monetary aggregates (NM0, NM1, NM2, NM3) and three measures of liquidity aggregates (L1, L2, L3) in India.
    a) NM0 is the monetary base.
    b) The NM1 is the non-interest bearing monetary liabilities of the banking sector
    c) NM2 represents the transactions balances of entities.
    d) NM3 includes NM2, long-term time deposits of residents and the call funding that the banking system obtains from other financial institutions.
2. Bank credit affects directly the consumption and capital formation. Provision of adequate flow of credit to the productive sectors is one of the important objectives of the monetary policy of RBI. The bank credit to the commercial sector includes advances in the form of loans, cash credit,

overdrafts, bills purchased and discounted, and investments in approved securities other than government securities. Further, in recent years, commercial banks are investing in various securities of the commercial sector like commercial paper, shares and debentures.

3. Net foreign exchange assets (NFA) of the banking sector include net foreign exchange assets of RBI and foreign assets of other bank. The latter include banks holdings of foreign currency assets net of i) their holdings of non-resident repatriable foreign currency fixed deposits which is presently defined to include FCNR(B) deposits, and ii) overseas foreign currency borrowings.

**Check Your Progress 1**

1) State the various aggregate measures of money supply in India.

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2) Explain the differences between NM0, NM1, NM2 and NM3.

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3) What are the liquidity aggregates in India?

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## 15.3 MONEY MULTIPLIER

The commercial banks in an economy operate on the basis of fractional reserve system. Suppose you deposit Rs. 1000 in a bank. The bank has to deposit a fraction of it with the central bank (In India, for example, the cash reserve ratio is 4.5 per cent in 2022; thus the bank has to deposit Rs. 45 with the reserve bank of India). Further, the bank has to retain certain percentage in the form of cash, gold and securities. This percentage is called statutory liquidity ratio (SLR). The SLR as of 2022 in India is 18 per cent, so that the bank has to maintain a SLR of Rs. 180. The bank knows that all the customers will not withdraw their savings at the same time. Therefore, the bank extends credit to its customers out of the remaining amount, i.e., 775. When individuals receive Rs. 775 as loans, and spend it, there is an increase in income of some individuals to the extent of Rs.

775. These individuals deposit part of this Rs. 775 in banks. The bank, on receiving the deposit, is in a position to extend further loans, and the process goes on.

Thus, the monetary base expands to an effective money supply through bank lending. The excess of the deposits of the general public over bank reserves is lent out in the form of bank loans, or government or corporate bonds. The non-bank public then deposits a fraction of these loans on checking accounts. Next, the banks lend out a fraction of these and so on. This process is called the money multiplier process. The ratio of the money stock measured as M1 to the monetary base, M0, is defined as the money multiplier, which is also equal to the inverse of the 'reserve ratio' or 'cash reserve ratio (CRR)'. Let us assume that the required reserve ratio is 25 per cent of total deposits (= demand deposits +time deposits, which is the liabilities of the banks) by public with the banks. So, the money multiplier is $\frac{1}{0.25} = 4$.

### 15.3.1 Mathematical Derivation of Money Multiplier

For mathematical representation, let us assume

C = currency held by the general public,

D = demand deposits held by the general public,

$C/D = cd,$ the desired currency deposit ratio (It implies, C = *cd* D)

R = bank reserves = currency held by the commercial banks plus their demand deposits in the central bank,

$R/D = rd;$ the required reserve ratio (It implies, R = *rd* D)

It is important to mention that *cd* is chosen by the non-bank public, whereas *rd* is determined by the central bank in the form of CRR and excess reserves in India. The reserve ratio is meant to maintain liquidity in the economy. In addition, banks may hold excess reserves according to their requirements of liquidity and lending risks.

Now, M1 = C + D = *cd* D + D = (*cd* +1)D … (15.1)

Monetary base = M0 = C + R = *cd* D + *rd* D = (*cd* + *rd*)D

Therefore, D = $\frac{M0}{cd+rd}$ … (15.2)

Substituting the value of D from (15.2) in (15.1), we obtain

$M1 = \frac{cd+1}{cd+rd}$ M0 = $m$M0

where *m* = $\frac{cd+1}{cd+rd}$. …(15.3)

The term *m* in equation (15.3) is the 'money multiplier'. It indicates the multiple by which money supply will increase, given the monetary base of the country. Suppose, for example, *cd* = 0.4 and *rd* = 0.1. In this case the money multiplier would be *m* = 2.8.

There are four implications of equation (15.3).

a) Money multiplier will be higher if we consider a broader measure of money supply (say, M3 in place of M1).

b) It is important to mention that $m$ depends upon the value of currency-deposit ratio ($cd$) and reserve ratio ($rd$). Any change in the value of these two parameters will change the value of $m$. The parameters $cd$ and $rd$ depend on various factors such as expected returns, degree of liquidity, and risk on alternative assets from the customers and bank's perspectives.

c) We can say that $\Delta M1 = m\Delta \text{M0}$. If we know the value of $m$, we can estimate by what extent M0 should be increased for a desired level of M1.

d) For given values of M0 and $cd$, the money supply will decrease with higher reserve-deposit ratio ($rd$) and vice versa. Now you can relate how changes in CRR and SLR influence money supply. An increase in CRR and SLR will decrease money multiplier. With the same monetary base, for implementation of a restrictive monetary policy the RBI will increase the reserve deposit ratio.

Money multiplier is generally stable under normal periods, but not always. For instance, during the early part of the Great Depression (1929 to 1933), there was significant decline in the value of the money multiplier in the US. Hence, depositors became nervous about the financial condition of banks, and withdrew their deposits. It led to an increase in $cd$; simultaneously it compelled banks to hold more reserves which increased $rd$.

### 15.3.2 An Alternative Interpretation of Money Multiplier

You may recall that money multiplier is defined as the ratio $\frac{M1}{M0}$. From equation (15.1) we know that M1 = ($cd$ +1)D . Now let us denote the public's desired currency-money ratio as $cm = \frac{C}{M1}$. By substituting the value M1= ($cd$ +1)D from equation (15.1), we obtain

$cm$ = C/($cd$ + 1)D = (C/D)/($cd$ + 1)

Thus we have

$$cm = \frac{cd}{cd+1} \qquad \ldots(15.4)$$

$$\text{and, } (1 - cm) = 1 - \frac{cd}{cd+1} = \frac{1}{cd+1} \qquad \ldots(15.5)$$

Substituting the above values in (15.3), we obtain

$$M1 = \frac{cd+1}{cd+rd}\text{ M0} = \frac{1}{\frac{cd}{cd+1}+\frac{rd}{cd+1}}\text{M0} = \frac{1}{cm+rd\,(1-cm)}\text{M0} = \frac{1}{1-(1-\text{rd})(1-\text{cm})}M0$$

…(15.6)

The monetary base is controlled by the central bank through variations in CRR and SLR. Apart from the above, there are channels such as open market operations, interest rates, and exchange rates.

To understand the mechanism of the effect of open market operations on the money supply, let us assume that the RBI increases M0 by the amount ΔM0 through open-market operations such as buying bonds. A seller of the bonds keeps the fraction *cm* as cash and deposits the remaining portion, (1– cm), at the bank. At the second round, the bank keeps the fraction *rd* of (1– cm)ΔM0 as reserves. The remaining amount, i.e., (1 – *rd*)(1 – *cm*)ΔM0 is given as loans to customers or buys bonds from the RBI. The money supply will be further increased by $(1-cd)^2(1-cm)^2\Delta \mathrm{M0}$ in the third round, and so on.

If *cm* and *rd* are constant (for simplicity), the total increase in money supply at the end would be:

$$\Delta M1 = \Delta \mathrm{M0} + (1-\mathrm{rd})(1-\mathrm{cm})\Delta \mathrm{M0} + (1-\mathrm{rd})^2(1-cm)^2\Delta \mathrm{M0} + \cdots = \frac{1}{1-(1-\mathrm{rd})(1-\mathrm{cm})}\Delta M0 \quad \text{... (15.7)}$$

(Equation (15.7) can be seen as an infinite geometric series. We apply the algebraic rule of sum of geometric series.)

We can conclude that an increase in monetary base by a specific amount leads to an increase in the money supply by *m* times of the increase in the monetary base.

**Check Your Progress 2**

1) Define money multiplier. Derive the money multiplier in terms of desired currency-deposit ratio and desired reserve-deposit ratio.

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2) What is the non-bank public's desired currency-money ratio? Explain how it is related to the money multiplier.

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## 15.3 OPEN MARKET OPERATIONS

As we have discussed earlier, there are four main functions of a central bank.

1. They act as the banker's bank or 'lender of last resort'. They are responsible for maintaining solvency and liquidity of the commercial banks.

2. They promote stable economic growth.
3. They act as the stabiliser of the purchasing power of the currency, or maintain stable inflation.
4. They manage the exchange rate in the market to reduce the volatility.

While these functions are complementary, there could be situations when one function conflicts with another. For instance, controlling of inflation can be sometimes achieved at the cost of economic growth. Thus when the objective is 'inflation targeting', economic growth may be somewhat lower. Similarly, in times of crisis the central bank raises the money supply to maintain liquidity in the economy and it creates inflationary pressure. The central banks work to achieve certain goals mainly targeting the interest rates, money supply targets and exchange rates, which are done through the monetary policy such as Open Market Operations (OMO), changes in reserve requirements, or changes in the bank rate.

Let us focus on the OMO. This is a popular policy tool available to central banks to influence the monetary base. Through this policy, the central bank buys government securities from the commercial banks and other financial institutions with an objective of increasing the monetary base. Such a policy stance is called expansionary monetary policy. The central bank makes payment for these securities in the form of reserves credited to the account held by the commercial banks with the central bank. It leads to a change in the composition of assets of the balance sheet of the commercial bank. It lowers the income producing assets like securities and increases the non-income producing assets like excess reserves of the commercial banks. Then, the commercial banks would try to convert a part or all of these excess reserves to provide new loans by lowering its interest rate. The central banks adopt expansionary monetary policy in a situation of higher demand for liquidity to keep the interest rates in check.

In contrast, the central bank opts for a contractionary monetary policy if its objective is to reduce the monetary base. In this case, the central bank sells the government securities to the commercial banks or other institutions to siphon-out some money from circulation. This process of increasing or reducing the monetary base by using the expansionary (contractionary) monetary policy, through buying (selling) of the government securities is known as the Open Market Operations. Recall that change in monetary base leads to change in money supply through the money multiplier.

**Box 15.1: Repo Rate and Bank Rate**

Most of the time we are confused between various interest rates such as bank rate or discount rate, repo rate and neighbouring rate in the banking system.

**Repo Rate:** The banks approach the central bank for financial assistance if they face crisis. The interest rate charged on banks by RBI to provide such financial assistance against of certain securities or bonds, is the repo rate. In other words, "commercial banks borrow money from RBI by selling securities or bonds with an agreement to repurchase the securities on a certain date at a predetermined price. The rate of interest charged by the central bank on the cash borrowed by commercial banks is called the Repo Rate". For example, if a commercial bank borrows or takes a loan of Rs. 100,000 at the predetermined repo rate of 5%, the interest to be paid to RBI at the maturity is Rs.5000.

In other scenario, if a commercial bank has excess funds, the bank deposits it in RBI and earns interest at a rate known as 'reverse repo rate'. For example, if a commercial bank deposits Rs. 100,000 at the reverse repo rate of 4%, the interest to be paid by RBI to the bank at the maturity is Rs.4000.

RBI controls the liquidity in the market through repo rate. With an objective of increasing liquidity, RBI reduces repo rate, which encourages the commercial banks to sell their securities and borrow from the RBI. On the other hand, RBI increases the repo rate, which discourages the banks from selling their securities, which decreases liquidity or the supply of money.

**Bank Rate:** The rate of interest charged by the central bank on the loans it has extended to commercial banks and other financial institutions is called "Bank Rate". In this case, there is no repurchasing agreement signed, no securities sold, and no collateral involved.

Banks also borrow funds from RBI and extend loan to their customers, which is an important source of their profit. Hence, they borrow funds from RBI without involvement of the repurchasing agreement, securities or bonds. The interest charged on such borrowing of commercial bank is the bank rate, which is usually higher than Repo Rate. This is an important weapon of RBI to control liquidity in the market. The increase in bank rate by RBI increases the borrowing costs of the banks which, in turn, reduce the supply of money. The bank rate is also called the discount rate.

There is another concept in the banking system that is the overnight rate. The commercial banks also borrow funds among themselves. The rate charged on such borrowing of funds is known as the overnight rate.

Sometimes the OMO is used to neutralise the impact of other factors on money supply. For instance, change in the exchange rate affects money supply. The central bank intervenes in the foreign exchange market by buying or selling foreign exchange to regulate the volatility of the domestic currency. Suppose, for example, the objective of the RBI is to check the appreciation of the Rupee. In order to realise this objective, the RBI would buy foreign currency (say US Dollar). With supply remaining the same, if the demand for US Dollar increases, the value of Indian Rupee will decrease. In the process, there would be some increase in money supply. In order to neutralise the increase in money supply, the RBI may choose the option of open market operations, and sell government securities in the market. It means that a central bank sells government securities to absorb the extra money supply, which came up due to its exchange rate policies. Neutralising such impact of change in foreign exchange reserves on the money supply through the OMO, or the selling of government securities, is known as the process of 'sterilization'.

Specifically, if the goal is to control inflation, the central bank sets a target of a higher interest rate by selling securities in the open market. A higher interest rate indicates less liquidity in the banking system, which leads to an increase in the interest rates charged by banks to their customers. Along with the decline in consumer expenditure, it also makes certain investment projects costly. As a consequence of higher interest rate, investment spending in the economy will decline. As a result of higher interest rate, consumption expenditure may decline, particularly those segment which is funded by borrowings from commercial banks. Further, households may allocate more to saving and less to consumption because of the increase in interest rate. As these are the components of aggregate demand, aggregate demand would decline because of the rise in interest rate. If aggregate demand falls, the price level would be negatively affected and the intended goal of controlling inflation is realised.

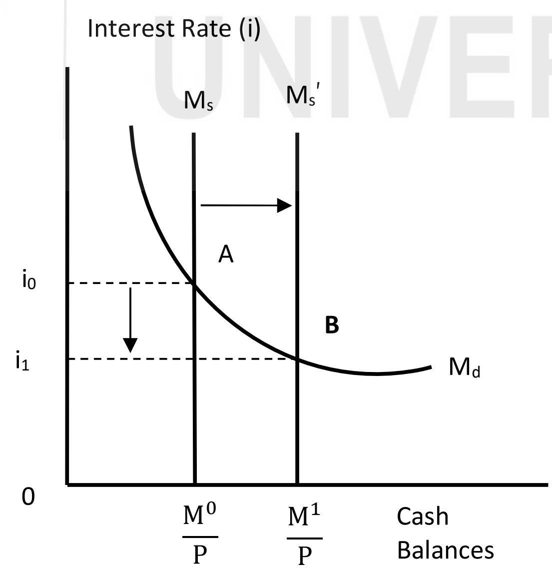


**Fig.15.1: Effect of Increase in Money Supply**

Let us explain the effect of an increase in interest rate on money supply through diagram. In Fig. 15.1 we indicate cash balances, money supply and money demand on the x-axis. On the y-axis we take interest rate ($i$). We know that demand for money depends upon income and interest rate such that, Md = f[Y, i]. Here demand for money is an increasing function in income (Y) and decreasing function in interest rate. Money supply is given by a vertical straight line at the level Ms = M/P, where M/P indicates the *purchasing power* of available money balances. In equilibrium,

$$M_d = M_s \text{ or } M_d = M/P \qquad \ldots(15.8)$$

If the price level remains unchanged, the real cash balances can be changed only through the changes in money supply. An increase in money supply or real money balances through OMO tools of the central bank shifts the $M_S$ function to the right (from $M_s$ to $M_s'$ in Fig.15.1). The main goal of such a policy is to stimulate aggregate demand. The implementation of this policy results in excess supply of money mainly in the form of excess reserves and thus loanable funds/ reserves. Consequently, it results in a fall in the interest rates charged by the commercial banks to customers on credit.

**Check Your Progress 3**

1) What is meant by open market operations? Explain how it affects money supply and interest rate in the economy.

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2) Explain the differences between bank rate, discount rate, repo rate, reverse repo rate, and overnight rate.

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## 15.4 LET US SUM UP

There is no unique definition of money supply. Central banks measure money supply through a series of monetary measures. The currency notes and coins, and the deposits of banks with the central bank form the monetary base of an economy. Monetary aggregates such as M0 and M1 are known as 'narrow money' whereas M3 is said to be 'broad money'.

The monetary base of an economy increases over time, which increases the money stock in the economy in a multiplicative manner. The multiple by which money stock increases against the monetary base is known as money multiplier. Money supply can be controlled through various tools of the central bank such as open market operations, changes in the rate of interests and exchange rates. If the objective of the central bank is to check inflation, the central bank would regulate money supply by buying government securities from the commercial banks and other financial institutions. The RBI can also increase the bank rate and repo rate, which will increase the borrowing cost of commercial banks, which in turn would decrease money supply. Higher interest rate affects investments and household consumptions negatively. It leads to decline in the aggregate demand in the economy which affects the price level. That means there is a strong linkage between the monetary policy of an economy and its real sector.

## 15.5 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

**Check Your Progress 1**

1) Refer to Section 15.2 and Table 15.1. Your answer should include the monetary aggregates NM0 to NM3, and L1 to L3.

2) Refer to Section 15.2, particularly Table 15.1, and answer.

3) Refer to Section 15.2, particularly Table 15.1, and answer.

**Check Your Progress 2**

1) It is the ratio of the money stock measured as M1 to the monetary base. Derive the equations on the basis of Section 15.3.

2) It denotes the ratio of currency with the non-bank public to the money stock measured by M1. Derive equation (15.6) and interpret it.

**Check Your Progress 3**

1) The process of changing the monetary base through buying or selling of the government securities is known as the Open Market Operations. Use Fig.15.1 to explain the effect of interest rate on money supply and interest rate.

2) Refer to Box 15.1 in Section 15.3 and answer.

# UNIT 16 CONDUCT OF MONETARY POLICY*

**Structure**



## 16.0 OBJECTIVES

After going through the unit you will be in a position to:

- distinguish between goals and targets of monetary policy;
- explain various credit control measures under monetary policy;
- identify the instruments of monetary policy and their role; and
- identify and explain the monetary transmission channels.

## 16.1 INTRODUCTION

In Unit 5 of this course we learnt that in an open economy with flexible exchange rate, monetary policy is effective in bringing about economic growth. You can appreciate the importance of monetary policy in view of the fact that most economies these days follow a flexible or floating exchange rate.

You may recall that the Classical Economists assumed money to be neutral in the sense that it does not have any effect on prices and economic growth. They believed that there is duality in the economy such that monetary and real variables do not influence one another. It was Keynes who suggested that money supply affects real variables. According to him increase in money supply leads to a decline in rate of interest, which in turn leads to greater borrowings by firms thereby leading higher investment. Higher investment leads to higher output and income of the country through a multiplier effect. Thus he visualized a greater role for the government. According to him economic policy should be countercyclical – it should neutralize the effects of business cycles. The New Classical economists attempted to justify the Classical views on the neutrality of

* Prof. Kaustuva Barik and Ms. Apica Sharma, Indira Gandhi National Open University, New Delhi

money and brought forth the 'policy ineffectiveness proposition'. According to the New Classical economists, it is best for the government to stick to certain 'policy rules'. Governments in all the countries, however, see an important role for monetary policy in the economy and actively pursue a monetary policy.

In this Unit we will discuss the types of monetary policy that a government pursues, the objectives of monetary policy, and the policy instruments available to policy makers. In addition, we will discuss how monetary policy works. We will bring out certain important channels; the effect of an increase in money supply on other economic variables.

## 16.2 MONETARY POLICY TYPES

The macroeconomic policy set by the central bank or the monetary authority of a country is known as monetary policy. It is concerned with the management of money supply in an economy. The change in money supply usually results in a change in rate of interest. The objective of monetary policy is to fulfill the objectives of inflation, growth, consumption and liquidity in the economy. Monetary policy also aims at maintaining exchange rates with other currencies in order to benefit trade, international confidence and policy stability in the country. There are certain instruments of monetary policy as we will learn later in this Unit.

Monetary policy could be of two types: expansionary and contractionary. With an expansionary monetary policy, the monetary authority tries to stimulate the economy using the policy instruments. In contrast, the monetary authority attempts to restrict money supply and control inflation with a contractionary monetary policy. In an expansionary monetary policy, either a lower than usual rate of interest is maintained or money supply is abnormally increased. This is done with the hope of providing cheaper credit to the firms so that investment in the economy increases. Increase in investment is expected to further increase employment and output. Expansionary monetary policy, however, often leads to deterioration of the exchange rate of the country with other currencies. If increases in money supply lead to higher inflation in the economy, depreciation of the domestic currency in the foreign exchange market may take place.

Under contractionary monetary policy, the instruments of monetary policy aim at a different set of results. If the economy is passing through a very high inflationary condition (if the economy is over-heated), the monetary authority usually resorts to a higher than usual rate of interest or abnormally low money supply. If contractionary monetary policy is implemented too vigorously, there may be decrease in production, increase in unemployment, and fall in aggregate demand leading to recessionary stage of the business cycle.

## 16.3 GOALS AND TARGETS OF MONETARY POLICY

You should note that there is a difference between the *goals* and the *targets* of monetary policy. Goals of monetary policy are the objectives it aims to achieve, that is, reducing inflation, achieving full employment or maximising economic growth. On the other hand, targets of monetary policy are the variables targeted such as money supply, bank credit, and short term interest rates through the instruments of monetary policy.

Monetary policy is one of the instruments of the overall economic policy of a country; hence its objectives are no different than the overall economic policy objectives of the economy. As per the prevailing economic, political, historical and international conditions, there are changes in the priorities given to the objectives of the monetary policy. Hence, they differ amongst developed and developing countries. The significant objectives of monetary policy are

- to ensure price stability, that is, containing inflation,
- to promote economic growth,
- to ensure stability of the exchange rate of the currency,
- to achieve full employment,
- to reach equilibrium in the balance of payments account, and
- to reduce economic inequality.

Let us explain below these objectives in some detail.

**a) Price Stability**

Controlling inflation is generally considered to be the foremost objective amongst developing economies as high inflation has an otherwise negative impact on growth and distribution. Price stability means relative price stability, depending on the prevailing conditions in the economy. Hence, by maintaining relative price stability there is a considerable check on the fluctuations in the business cycle leading to stability in the economic growth process as a whole.

Price stability does not mean that prices should not increase at all. There should be a reasonable rate of inflation in an economy so that there is incentive for production and economic growth. There is no consensus among economists, however, on an ideal rate of inflation; it varies across countries and over period of time. In countries like the US, an inflation rate of around 2 per cent is targeted. In India, an inflation rate of 4 per cent with a tolerance band of 2 per cent is considered as ideal (it implies that inflation should not be outside the range of 2 per cent to 6 per cent).

Many countries have faced episodes of abnormally high inflation. In 2019, for example, some countries are experiencing exceptionally high inflation – Venezuela has an inflation rate of about 10 million per cent per year! Argentina, Iran and Sudan are facing inflation rate of above 25 per cent per year. Abnormally high inflation leads to rise in the cost of living of the poor, makes exports more expensive, lowers the rate of savings which further hampers economic growth and encourages unproductive investment such as jewelry, real estate, etc. During abnormally high inflationary scenario, contractionary monetary policy is used. In period of relatively stable inflation, the monetary authority pursues a monetary policy such that price stability with economic growth and employment generation are maintained.

**b) Economic Growth**

Promoting economic growth is another important objective of both developed and developing countries. Economic growth implies growth in the productive capacity of the economy so that real national income increases. To achieve economic growth, the rate of savings and investments need to increase. Monetary policy can help produce favorable effects to the requirements of economic growth by directing the banking sector towards priority sector lending or making appropriate changes in interest rate to incentivize people to save.

With the openness of the economy and floating exchange rate, the goal to achieve increased economic growth through instruments of monetary policy may contradict with the goal of exchange rate stability. In order to prevent the depreciation of the currency, a tight monetary policy need to be maintained but the higher rate of interests will reduce liquidity in the banking system. This reduction in liquidity results in increased cost of lending and lesser production and increased unemployment in the economy. These contradictory objectives result in a trade off in the policy circles and considering the other political, social and economic objectives, a final decision is reached.

**c) Exchange Rate Stability**

Almost all major economies of the world have started following the flexible exchange rate system under which the exchange rate of the currency is determined by the demand for and supply of foreign exchange. Whenever there is disequilibrium in the demand and supply of foreign exchange, the external value of the currency changes. The domestic investors demand foreign currency when there is profitable investment possible abroad whereas they demand home currency more when there is profitable investment at home.

Under floating exchange rate the central bank or the monetary authority intervenes to avoid sharp fluctuations in the exchange rate. In order to prevent depreciation of rupee, the central bank can release more foreign currency from its foreign exchange reserves. Also the central bank can increase the short term lending rates and increase the required cash reserve ratio (CRR) in order to reduce the demand for the foreign currency.

**d) Full Employment**

In order to achieve maximum social welfare it is imperative that the economy attains full employment. Any country will prosper when all its people get the job that they desire to do at the prevailing wage rate. With a tight monetary policy, the objective of full employment and growth contradicts. Hence, in order to increase the level of employment the central bank or the monetary authority will have to adopt expansionary monetary policy.

**e) Balance of Payments Equilibrium**

Abnormally high current account deficit (CAD) is financed through increase in nominal capital inflows which is achieved through the depreciation of the home currency. Thus, imports become very expensive and there is increase in inflation in the economy. A tighter monetary policy helps reduce these effects through high cost of credit and lower availability of money with the importers, thus curbing imports.

Hence, monetary policy plays a significant role in maintaining the current account deficit at the reasonable level. Balance of Payments equilibrium implies exchange rate stability.

**f) Economic Inequality**

There is increased income inequality amongst the developed and developing countries. This increased inequality leads to reduction in social welfare and increased frustration amongst the masses. Monetary policy as an economic instrument aims to achieve equitable distribution of income and wealth through priority sector availability of credit to the weaker sections of the society at very low rate of interests.

The targets of monetary policy are generally the following three parameters:

- to regulate the supply of money/ credit
- to maintain a reasonable rate of interest or cost of money/credit
- to conveniently avail money/ credit

**a) Supply of Money**

The currency issued by the monetary authority and the demand deposits with the commercial bank comprise the supply of money. The issue of currency by the central bank needs to be flexible enough depending on the inflationary or recessionary stages prevailing in the economy. The banking system also needs to play a progressive role in facilitating demand deposits with the banks.

**b) Rate of interest or Cost of Money**

Underdeveloped economies generally adopt an expansionary monetary policy in order to promote sectors like agriculture and industry. Policy of discriminatory rate of interests is often followed in order to promote priority sector lending. For example, lower rate of interest is charged from the priority sectors such as micro, small and medium enterprises (MSME). Simultaneously, higher rates of interest may be charged from unproductive and luxurious spending so that higher credit is made available to the priority sector.

During inflationary stage of the business cycle, the monetary authority keeps the rate of interest at a higher level to curb money supply in the economy. In contrast, during the recession phase, the monetary authority may decide to keep the interest rate at a lower rate. It is expected that lower interest rate will provide incentives for higher production, employment and growth.

**c) Availability of Credit**

The ease with which credit is available in the economy is an important factor for economic growth. The eligibility and norms to avail credit from the financial system are set by the central bank or monetary authority of the country; hence, these norms are part of the monetary policy.

## 16.4 MONETARY POLICY INSTRUMENTS

The various instruments of monetary policy to control credit are divided in two categories, namely, Quantitative and Qualitative measures. Quantitative credit control measures are non-discriminatory in nature, say for example, when a certain interest rate is set by the central bank of a country, that rate applies to the banking system of the country as a whole. In contrast, Qualitative / Selective credit control measures vary from one section of the society to the other.

### 16.4.1 Quantitative Credit Control Measures

The important quantitative credit control instruments of the monetary policy are as follows:

- Bank Rate
- Open Market Operations
- Change in Minimum Reserve Ratio
- Change in Liquidity Ratio

**a) Bank Rate**

The most noticed and significant instrument of monetary policy is the rate of interest on borrowing and lending set by the central bank or monetary authority of the country. Bank rate is the rate of interest at which the central bank provides loans to commercial banks and other financial institutions. Increase in bank rate has the effect of increasing the rate of interest in the economy. Similarly, decrease in bank rate lowers the rate of interest in the economy. Higher bank rate lowers the extent of credit creation in the economy which leads to a decline in aggregate demand and hence lower prices. On the other hand, in a recessionary phase a lower bank rate is propounded. Borrowing from the central bank, usually for short period, by the commercial banks is done through the 'repo rate'. Similarly, commercial banks can deposit their excess liquidity in the central bank for which the 'reverse repo rate' is applicable.

**b) Open Market Operations**

Purchase and sale of securities in the open market by the central bank or the monetary authority is popularly known as open market operations. In order to contract the credit in the economy, the central bank sells securities in the open market. This leads to fall in aggregate demand and reduction in price level. Whereas, when credit is to be expanded, there is purchase of securities by the central bank in the open market. This leads to increase in money supply which leads to increase in aggregate demand and production level in the economy.

**c) Cash Reserve Ratio**

The commercial banks are required to keep a certain percentage of their total deposits as cash with the central bank of the country. If the aim of the central bank is to reduce supply of money in the economy, it will raise the cash reserve ratio (CRR). An implication of higher CRR is that commercial banks are required to keep a higher percentage of their deposits in the form of cash with the central bank. Thus a lesser amount is left for lending purposes. In contrast, reducing the CRR would increase the credit creation in the economy as commercial banks have larger availability of funds with them for the purpose of lending and creating credit.

**c) Statutory Liquidity Ratio**

Statutory liquidity ratio (SLR) requires every bank to keep certain proportion of its total deposits in the form of liquid (or near liquid) assets. When the central bank needs to reduce credit, it raises the SLR. On the other hand, when the objective of the central bank is to expand credit, it reduces the SLR.

### 16.4.2 Qualitative Credit Control Measures

Very often the monetary authority in a country controls the supply of credit in the economy. Thus financial institutions may not extend credit to a borrower even if he/she has a demand for credit. These measures are selective in nature (because of it, they are often called selective credit control measures also) such that these measures target certain sectors of the economy or certain sections of the society. Secondly, it is discriminatory in the sense that different segments get differential treatments. Thirdly, there is a market failure in the sense that supply and demand for credit are not taken into consideration. Fourthly, there is disequilibrium in the credit market – supply and demand do not match.

We can classify qualitative nature can be classified in the following types:

- Change in margin requirement
- Credit rationing
- Direct action

**a) Margin Requirement**

Margin requirement is the margin between the value of goods pledged as security and the amount of loan. If the central bank finds that traders of any particular area are stock piling certain commodities which leads to an increase in prices of those commodities in the market, then the margin requirements of those commodities are increased. This way credit is contracted and those defaulting traders are punished.

**b) Credit Rationing**

May a time the central bank resorts to credit rationing. Through rationing of credit, the central bank can perform the following tasks:

- It can decline loan to a particular commercial bank
- It can reduce the total amount of loan to all the commercial banks in general.
- It can fix a certain quota for different banks.

Hence, through the above selective credit rationing methods, the central bank can increase or reduce the credit in the market.

**c) Direct Action**

Central bank can take extreme actions against those commercial banks that disobey its directives. One such extreme action is directly imposing harsh restrictions on the credit of those commercial banks.

## 16.5 MONETARY POLICY IN INDIA

Reserve Bank of India (RBI) is the institution responsible for monetary policy framework in India. The policy targets and instruments of RBI have changed over the years in tandem with the domestic and global structural changes. Hence, in brief we discuss the evolution of monetary policy in India.

### 16.5.1 Early Years of Monetary Policy

Post Independence; say from 1950 to 1970, the RBI provided support to the government's plan expenditure using deficit financing. It was the industrialization phase when large scale public sector investments were financed by borrowing from the RBI. The RBI then focused on selective credit control check inflation while making sure that credit flowed to priority sectors. Since 1971, up to mid 1980s, the Indian government relied more on fiscal policy measures compared to monetary policy as Statutory Liquidity Ratio (SLR) and Cash Reserve Ratio (CRR) were increased. The high level of inflation in the 1980s led to the adoption of monetary targeting framework on the recommendations of the Sukhamoy Chakravarty Committee in 1985.

After the economic reforms of 1991, the RBI's role underwent a significant change. Steps were taken to curb the monetization of fiscal deficit. Say for example, by 1997 the ad-hoc treasury bills were completely phased out. However, monetization of fiscal deficit continued through primary issue of public debt in auctions. Since April 2006, the RBI is not allowed to subscribe to the primary issue of public debt.

### 16.5.2 Liquidity Adjustment Facility

Multiple indicator approach was adopted by RBI in 1998 in which high frequency financial market rates and fortnightly/ monthly indicators were tracked along with the output data for monetary policy purposes. In April 1999, on the recommendations of the Narasimham Committee (second committee), Interim Liquidity Adjustment Facility (ILAF) was introduced. Repo and reverse repo rates emerged as the primary policy rates of the RBI and the significance of CRR and SLR as instruments of RBI to manage liquidity got reduced. However, the ILAF suffered from non-existence of a ceiling rate and no unique policy rate. Hence, since May 2011 the repo rate has become the only independently varying rate under the Revised Liquidity Adjustment Framework (RLAF). In September 2014 the RLAF was fine-tuned following the recommendations of the Urjit Patel Committee. In May 2016, the Monetary Policy Framework Agreement (MPFA) was signed between the Government of India (GOI) and the RBI to regulate inflation and bring it below 4 percent in financial year 2016-17. In case of failure on the part of the RBI to comply with the above targets, it is accountable to the GOI and suggest reasons and remedies to achieve the same in future by a specified date.

Since February 2019 the RBI has followed an accommodative monetary policy in order to provide stimulus to the ailing economy. The RBI, however, realized that the Scheduled Commercial Banks (SCBs) do not transmit the rate cuts to the general public. In view of the above, the RBI in August 2019 decided to formalize the linkage between repo rate and bank lending rates.

## 16.6 TRANSMISSION MECHANISM OF MONETARY POLICY

As mentioned in the beginning of the Unit, monetary policy affects several key economic variables in the economy. Monetary policy decision, particularly on money supply and interest rate, sets in motion a sequence of effects. Transmission mechanism means the process or the channels through which monetary policy affects economic variables.

It is difficult to make a complete list of variables which get affected by a monetary policy decision. Economists have attempted to make a comprehensive list of the channels. There are differences amongst economists, however, on the importance of various channels. We have tried to compile the various channels (See Table 18.1) based on the studies of Mishkin (1995), Taylor (1995) and Boivin et al. (2010). Second, the impact of monetary policy on all variables is not uniform – while certain variables get influenced by a particular monetary policy decision to a great extent, the impact is negligible on certain other variables. It depends on the policy decision and its target variables. Third, some of these effects are visible immediately while certain others are noticeable with a lag.

**Fig. 18.1: Monetary Transmission Mechanism**

| CHANNEL | CAUSATION CHAIN |
|---|---|
| **Interest Rate Channel**<br>(a) Nominal Variant | Money ↑ → Interest Rate ↓ → Investment ↑ → Aggregate Demand ↑ |
| (b) Real Variant | Money ↑ → Expected Price ↑ → Expected Inflation ↑ → Real Interest Rate ↓ → Investment ↑ → Aggregate Demand ↑ |
| **Open Economy Channel**<br>Exchange Rate Channel | Money ↑ → Interest Rate ↓ → Exchange Rate ↑ (Depreciates) → Exports ↑ → Net Exports ↑ → Aggregate Demand ↑ |
| **Credit Channel**<br>(a) Bank Lending | Money ↑ → Deposits ↑ → Bank Loans ↑ → Investment ↑ → Aggregate Demand ↑ |
| (b) Balance Sheet | Money ↑ → Price of Stocks (Net Worth) ↑ → Adverse Selection ↓ → Moral Hazard ↓ → Lending ↑ → Investment ↑ → Aggregate Demand ↑ |
| (c) Cash Flow | Money ↑ → Interest Rate ↓ → Cash Flow ↑ → Adverse Selection ↓ → Moral Hazard ↓ → Lending ↑ → Investment ↑ → Aggregate Demand ↑ |
| (d) Unanticipated Price Level | Money ↑ → Unanticipated Price of Stocks ↑ → Adverse Selection ↓ → Moral Hazard ↓ → Lending ↑ → Investment ↑ → Aggregate Demand ↑ |
| (e) Liquidity Effects | Money ↑ → Price of Stocks ↑ → Value of Financial Assets ↑ → Likelihood of Financial Distress ↓ → Consumer Durables and Housing Expenditure ↑ → Aggregate Demand ↑ |
| **Other Channels**<br>(a) Wealth Effect | Money ↑ → Price of Stocks ↑ → Wealth ↑ → Consumption ↑ → Aggregate Demand ↑ |
| (b) Tobin's q | Money ↑ → Price of Stocks ↑ → Tobin's q ↑ → Investment ↑ → Aggregate Demand ↑ |
| (c) Expectations | Money ↑ → Expected inflation ↑ → Consumption/Investment ↑ → Aggregate Demand ↑ |

We now elaborate on some of the above transmission channels.

**a) Interest Rate Channel**

Nominal Interest Rate: When there is an increase in money supply it leads to a decline in nominal interest rate which further raises the incentives for investment. An increase in investment has obvious impact of increase in aggregate demand.

Real Interest Rate: An increase in money supply increases the expected price level, that is, it raises the rate of inflation in the economy leading to fall in the real rate of interest. The fall in interest rate raises investment and as a consequence the aggregate demand. The increase in expected price also provides an incentive to firms to increase production.

We know from rational expectations hypothesis that the average of short term rates is the long term rate. Therefore, transmission from short term rates to long term rates is essential for this transmission channel to be effective. For a developing economy like India, the costs of funding does not decrease in short term as substantial segment of the deposits in banks are long term in nature. Developing countries generally have an underdeveloped bond markets, less competition in the banking sector, and existence of a large informal sector for credit. These factors make the transmission mechanism through this channel extremely weak.

**b) Open Economy Channel**

Exchange Rate: An increase in money supply leads to fall in interest rate. Fall in interest rate leads to outflow of funds, as investors find it more profitable to invest abroad. This leads to decrease in supply of foreign currency which leads to depreciation of the domestic currency. Depreciation of domestic currency leads to increase in net exports leading to an increase in aggregate demand.

After the economic reforms of the 1990s, the Indian economy gradually opened up to the external sector. Therefore, the importance of this channel in monetary transmission in India is on the rise.

**c) Credit Channel**

Bank lending: An increase in money supply leads to an increase in bank deposits which enhance bank lending. Increase in bank lending leads to further investments; leading in turn to increase in aggregate demand.

Banks may be reluctant to lend despite the growing cash reserve due to depressed economic conditions, and some borrowers should be completely bank dependent in order to make this channel of transmission really effective. In countries like India, where financial markets are not fully developed as yet, this could be a significant channel of transmission. Balance Sheet: An increase in money supply raises the prices of the stocks as people invest money in stocks. This reduces the drawbacks of asymmetric information which boosts confidence of the lenders and borrowers on each other. There is increased lending and thus investments in the growing economy which ultimately raises the aggregate demand.

Cash Flow: An increase in money supply leads to fall in interest rate and increase in cash flow in the economy. This again reduces the drawbacks of asymmetric information and increased investment and aggregate demand follows.

Unanticipated Price level: An increase in money supply raises the otherwise unexpected rise in price level of stocks in the economy. The drawbacks of asymmetric information are reduced with further raises lending, investment and aggregate demand.

Liquidity Effects: An increase in money supply raises the prices of stocks and value of other financial assets. This reduces the risk of financial distress further leading to increased consumption of consumer durables and hence increased overall aggregate demand.

**d) Other Channels**

Wealth effect: An increased money supply raises the price of stock and thus raising the wealth of the people. This increased wealth raises consumption and aggregate demand.

Tobin's Q Effect: '**Q**' represents the ratio of the market value of a firm's existing shares (share capital) to the replacement cost of the firm's physical assets (thus, replacement cost of the share capital). An increase in the money supply raises the price of the stocks and thus raises the 'Q' value leading to increased investment and aggregate demand.

Expectations: An increased money supply raises the expectations of increased inflation in the economy which enhances present consumption and investment leading to increased aggregate demand.

## 16.7 LET US SUM UP

Monetary policy is the macroeconomic policy implemented by the central bank of the country in order to regulate credit through its various instruments. The credit control measures are classified into quantitative and qualitative. The quantitative measures are interest rates, open market operations, reserve requirements and liquidity ratio conditions. The qualitative measures are changes in margin requirements, rationing of credit and direct action. The transmission of monetary policy takes through various channels. These channels include interest rate channel, open market channel, credit channel and other channels such as expectations, Tobin's q and wealth effect. Every channel shows that an increased money supply through various channels ultimately increase the aggregate demand in the economy. In developed countries the dominant channel of transmission is the traditional interest rate channel, whereas in developing countries the exchange rate and bank lending channel play a significant role.

## 16.8 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

**Check Your Progress 1**

1) By monetary policy we mean a deliberate change in the interest rates and money supply by the central bank or monetary authority of the country.
2) The goals and targets of monetary policy are different and ultimately move towards the overall objective of economic policy in general.

**Check Your Progress 2**

1) There are two types of credit control measures of monetary policy, namely, Quantitative and Qualitative measures. The Quantitative measures are non discriminatory in nature and apply to the entire country, while the qualitative measures are selective or discriminatory in nature.

**Check Your Progress 3**

1) There are various channels of transmission of monetary policy.

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# UNIT 17 THEORY OF MONETARY OF POLICY*

**Structure**



## 17.0 OBJECTIVES

After going through this unit you would be able to:

- explain the discretion and rule of the Central Bank in determining inflation and output;
- understand the current consensus of monetary policy;
- assess the impact of a higher inflation target on interest rate and aggregate demand;
- assess the impact of an expansionary fiscal policy on interest rate, inflation and aggregate demand;
- explain the path dependence in macroeconomic outcomes.

## 17.1 INTRODUCTION

In this unit we focus on three main themes related to the conduct of monetary policy. First, we discuss the rules vs. discretion debate. Discretionary policy is arbitrary and often mis-placed (policy makers may follow an expansionary policy while the need is for a contractionary monetary policy). Further, discretionary monetary policy results in various types of lags (such as identification lag, decision lag and implementation lag). Thus, policy rules are preferred to discretionary measures by mainstream economists. However, there appears to be no consensus on the definition of rules. According to Friedman, Central Banks

* Dr. Vineet Kohli, Tata Institute of Social Sciences, Mumbai

should follow the rule of constant monetary growth over time. Other economists such as Robert Barro define the rules as a commitment to zero/low inflation policy. For most new-Keynesian economists, policy rules are shown in terms of equations that link the Central Bank's policy rate (i.e., the key lending rate of the Central Bank; repo rate, for example, in India) to deviations from inflation and output targets.

Secondly, we will discuss about the current consensus on monetary policy. In the consensus model, monetary policy plays a major role in fixing inflation rate at its pre-defined target. In other words, monetary policy can successfully provide the nominal anchor. Moreover, monetary policy, while successful in determining the rate of inflation, cannot influence the level of output, which is fixed at its natural (i.e., full employment) level. In this sense, the new consensus is simply a revised version of Friedman's idea that market forces by themselves deliver full employment and monetary policy is neutral. At best, Central Banks can hope to reduce fluctuations of output around its natural level by following an appropriate interest rate policy.

Finally, we will talk about the criticism of the new consensus model. The new consensus model assumes that natural output is fixed. Critics point out that this may not be so. High aggregate demand (injected through monetary or fiscal means) may increase actual as well as natural output levels and take the economy to a higher equilibrium. We will use the concepts of IS curve, expectations augmented Phillips Curve, rational expectations, and adaptive expectations. You are familiar with these concepts from the earlier Units.

## 17.2 RULES, DISCRETION AND DYNAMIC CONSISTENCY

As we saw in Unit 2, Keynes advocated active government intervention. When there is shortfall in effective demand, the state was expected to increase government expenditure, may be through a deficit budget. Similarly, when the economy is passing through inflation, the government should go for a surplus budget. Such variation in government expenditure necessitated certain discretionary policy measures by the government. The classical economists, on the other hand, had suggested minimum government intervention in the economy so that market forces (demand and supply) can operate efficiently. The new classical economists suggest certain policy rules on macroeconomic variables.

It may be useful to discuss Friedman's (1968) prescription of what monetary policy should seek to achieve. First, Friedman weighed in against the then popular belief in trade-off between inflation and unemployment. According to Friedman, no such trade-off exists in the long run and the attempts to push output above its 'natural' level will only yield higher rates of inflation. As we shall see in the next section, this part of Friedman's analysis plays a very important role even in the current consensus.

Secondly, Friedman believed that Central Banks do not use their discretion entirely judiciously. For example, the Federal Reserve in the US contracted money supply during the Great Depression. When Central Banks did not err in the wrong direction, they did so by moving too far, thereby, causing wide swings in money supply. Thus, Friedman proposed a 'publicly stated policy of steady rate of growth of money supply' for Central Banks. Moreover, the constant money growth rule is superior to other rules that targeted a certain price level or inflation rate. This is because 'we cannot predict accurately just what effect a particular monetary action has on the price level and, equally importantly, just when it will have that effect'. Most Central Banks and economists today do not share Friedman's confidence in the ability of monetary authorities to fix money supply. They also do not share his pessimism regarding inflation targeting. Where the two – Friedman and the current mainstream – do converge is in their shared belief in the superiority of rules over discretion.

### 17.2.1 Loss Function

There is a belief that while rules are superior to discretion, they may be dynamically inconsistent – having announced the rule, Central Banks may subsequently find it optimal to violate it. Barro and Gordon's work is an important contribution in this area. Let us look at their main contributions through a simple model. First, let us introduce the loss function of a Central Bank.

The main objective of monetary policy is inflation targeting but control of inflation should not be at the cost of economic growth and employment. We assume that the economic welfare of the country is the maximum when the economy is operating at its natural output level (i.e., full employment output level) and inflation is at the targeted rate. There is a loss of welfare if inflation is different from the targeted rate and actual output is different from the natural output level. If we consider both these issues, we obtain a *loss function* of the Central Bank, given by

$$L = x^2 + (y - y'')^2 \qquad \text{... (17.1)}$$

In equation (17.1), we denote inflation rate by $x$ and output by $y$. The notation $y''$ is the output target pursued by the Central Bank. Let us denote natural output level by $y'$. Natural output stands for the output level that will be produced when labour force is fully employed. Let us assume that the Central Bank sets its output target in excess of the natural output level, $y'$. The Central Bank dislikes any inflation and deviations of output from its target. Higher value of the loss function ($L$) implies greater loss or welfare cost suffered by the economy.

The Phillips Curve is given by the following equation:

$$x = x^e + e(y - y') \qquad \text{... (17.2)}$$

Equation (17.2) says that actual inflation ($x$) will exceed expected inflation ($x^e$) whenever actual output ($y$) exceeds natural output ($y'$). Objective of the Central Bank is to choose that combination of output and inflation that minimises its loss function. Obviously, any such combination must lie on the Phillips Curve. Thus, the Central Bank minimises its loss function subject to the constraint of equation (17.2). The Central Bank tries to reach its optimal solution by varying either the money supply or the rate of interest. We can add another equation to the model that relates aggregate demand (which equals output $y$) to money supply and inflation rate. The optimal combination of output and inflation can then be inserted in the aggregate demand equation to obtain optimal money supply. Without intending to carry out such an exercise, we simply assume that the Central Bank moves some monetary policy instrument under its control to reach its desired outcome.

The Central Bank's optimisation exercise is similar to that of a household in consumer theory. Just as the consumer tries to reach the highest possible indifference curve for a given budget constraint, the Central Bank tries to attain the lowest possible loss function for a given Phillips Curve. Fig. 17.1 shows the relevant picture. Loss functions of the Central Bank are given by concentric circles centred at B, where inflation rate is zero and output is at its target $y''$. The point B in Fig. 17.1 stands for the 'bliss point' as the Central Bank can do no better than to achieve its output target at zero inflation. The larger is the circle, the greater is the welfare cost incurred by the Central Bank. At each point on the vertical axis, output is at its natural level and, from the Phillips Curve equation, actual and expected inflation rates coincide. Since actual output increases with the inflation rate, the Phillips Curve is an upward-sloping line.

At an optimum point, the slope of the loss function is equal that of the Phillips Curve. Let the slope of the Phillips Curve be equal to $e$. The slope of the loss function is given by $e = \frac{(y''-y)}{x}$ (you can verify this by totally differentiating equation (17.1) and setting $dL = 0$). Hence, at the optimum point:

$$ex = y'' - y \qquad \text{... (17.3)}$$

After multiplying both sides of equation (17.3) by $e$ and adding with equation (17.2), we obtain:

$$x = [x^e + e(y'' - y')]/(1 + e^2) \qquad \text{... (17.4)}$$

In other words, the Central Bank's choice of inflation rate will vary positively with its output target, $y''$ and with the inflation rate expected by public, $x^e$.

### 17.2.2 Rules vs. Discretion

We now endogenise expected inflation rate by assuming that public's expectations are rational. That is, the public is aware that the Central Bank minimises its loss function subject to the constraint posed by the Phillips Curve and, thus, fixes inflation rate according to equation (17.4). Therefore, public will come to expect the inflation rate that the Central Bank actually fixes. This implies the following equation:

$$x = x^e = [x^e + e(y'' - y')]/(1 + e^2) \qquad \text{... (17.5)}$$

On solving the above equation, we obtain:

$$x = x^e = (y'' - y')/e \qquad \text{... (17.6)}$$

Equation (17.6) gives the discretionary outcome. Inflation rate under discretion is endogenous. It depends on the output target set by the Central Bank. A higher output target of the Central Bank results in a higher rate of inflation but fails to stimulate output above its natural level. This is because output can be stepped above its natural level only by increasing the inflation rate above its expected level. But the Central Bank cannot create any inflation surprise as long as the public forms its expectations rationally. When the Central Bank uses its discretion to step output above its natural level, all it manages to achieve is a positive inflation rate. In Fig. 17.1, the discretionary outcome is depicted at A. At this point two conditions are fulfilled: (i) slope of the loss function equals the slope of the Philips Curve, and (ii) actual inflation rate equals the expected inflation rate.

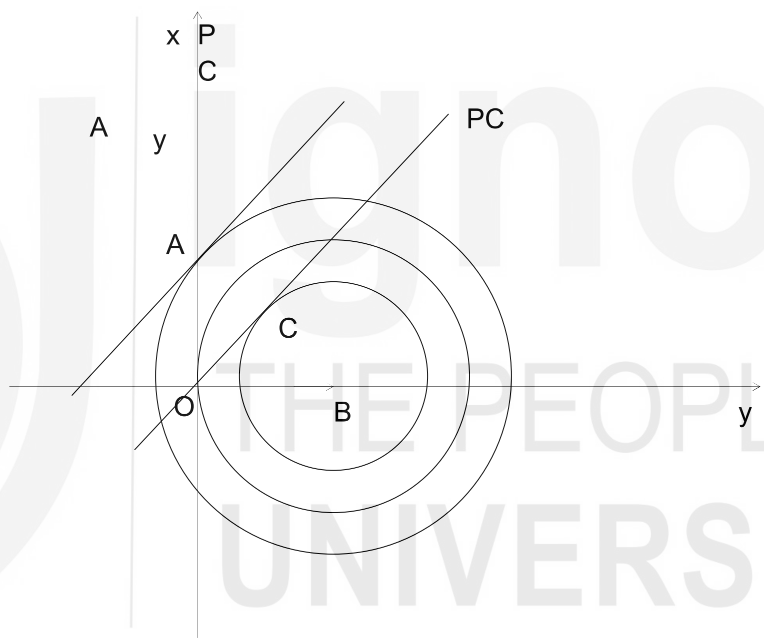


**Fig. 17.1: Rules vs. Discretion**

Now, suppose the Central Bank shifts from a discretionary policy framework to rule-based policy framework. Instead of arriving at certain positive inflation rate after solving its minimisation exercise, suppose the Central Bank declares a 'zero inflation rule'. Given rational expectations, the public will expect the rule that the Central Bank intends to increase production. Therefore, under the zero inflation rule:

$$x_r = x^e = 0 \qquad \text{... (17.7)}$$

where the subscript *r* stands for 'rule'.

### 17.2.3 Dynamic Inconsistency

Given our formulation of the Phillips Curve, equation (17.5) implies equality between actual and natural output. The rule-based solution produces the same level of output as the discretionary solution but at a lower inflation rate. Therefore, the 'zero inflation rule' implies a lower welfare cost than the discretionary solution. The rule-based solution is depicted by point O in Fig. 17.1. Now, suppose the Central Bank anchors expected inflation rate at zero by declaring a 'zero inflation rule'. Since actual inflation equals expected inflation on the vertical axis, the relevant Phillips Curve (PC) passes through O in Fig. 17.1. It is clear that the Central Bank can do better than point O if it *cheats* on its commitment by choosing point C on the PC line where its welfare cost is minimised. Since the Central Bank's minimisation exercise yields equation (17.5), we can find the relevant inflation rate by inserting $x^e = 0$ in equation (17.5). This gives:

$$x_c = e(y'' - y')/(1 + e^2) \qquad \text{... (17.8)}$$

where *c* stands for cheating.

Since $y'' > y'$, equation (17.8) implies a positive inflation rate. The intuition is simple. The Central Bank first anchors the public's inflationary expectations by announcing a 'zero inflation rate' policy and then creates an inflation surprise by violating its rule. Given our formulation of the Phillips' Curve, this causes the output level to rise above its natural level. Also, the Central Bank has convex preferences; it prefers averages to extremes. Therefore, the combination of zero output gap (excess of actual over natural output) and zero inflation rate entails a greater welfare cost than a point like C where both inflation rate and output gap are positive.

The discussion so far can be summed up as follows: Rules, though superior to discretion, suffer from dynamic inconsistency. Having announced a rule, the Central Bank would be tempted to violate it. The problem is that once the public's trust is broken, the public would come to believe that the Central Bank, regardless of what it proclaims, would apply its discretion in fixing the inflation rate. The Central Bank would then maximise its welfare by taking public's inflationary expectations as given. The Central bank may discard its rule-based policy and end up with the discretionary case. Such shift in policy, however, should be avoided; because such change would entail inflation with no gain in output.

Mainstream economists maintain that delegation of monetary policy-making to conservative Central Bankers, who place a high weight on inflation control, can help to address the dynamic inconsistency problem. The benefits of delegation to the conservative Central Banker were captured in a model developed by Kenneth Rogoff. Suppose the conduct of monetary policy is delegated to an individual whose loss function is given by:

$$L = \gamma x^2 + (y - y'')^2 \qquad \text{... (17.9)}$$

where $\gamma > 1$.

The new loss function places a higher weight on inflation. The conservative

Central Banker minimises the objective function given by equation (17.9) subject to the constraint given by the Phillips Curve (see, equation (17.2)). The public believes in rational expectations, and thus, expected and actual inflation rates are equal. The solution under discretion is now given by:

$$x = \frac{y'' - y'}{\gamma e} \qquad \text{... (17.10)}$$

The intermediate steps involved in derivation of equation (17.10) are skipped here. We leave it as an exercise for the reader (hint: repeat the steps that were earlier followed in deriving equation (17.6)). Since $\gamma > 1$, inflation rate falls after monetary policy is delegated to a conservative Central Banker (this can be verified by comparing equation (17.10) with equation (17.6)). The government, which delegates policy-making to a conservative Central Banker will benefit, since a lower rate of inflation also reduces the value of the government's loss function given by equation (17.1).

Of course, delegation of policy-making to a conservative Central Banker will work only when the banker is independent of government control. The government may not be as inflation-averse as the Central Banker (if it is as inflation averse, nothing is gained from delegation). Thus, delegation alone will not lower the inflation rate; the conservative Central Banker must be free from the control of the government to pursue the objective of inflation control.

**Check Your Progress 1**

1. Explain why policy rule is preferred to discretion.

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2. What is meant by dynamic consistency?

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## 17.3. CONSENSUS VIEW OF MONETARY POLICY

We start the analysis by describing the main objectives of a Central Bank. We hypothesize that the Central Bank has an inflation target of $x'$ in mind. It also wishes to keep output at its natural level, $y'$. The main instrument of the Central Bank policy is the rate of interest. In older versions of many textbooks, it is assumed that the Central Bank pursues its objectives by regulating the supply of money. Under this policy, the interaction between supply of money and demand for money determines the rate of interest. An implication of the above is that money supply is an exogenous variable (determined by the Central Bank) while interest rate is an endogenous variable.

In recent years, the consensus has shifted in the other direction. Most economists today argue that the main instrument of monetary policy is interest rate. For example, the Reserve Bank of India fixes its repo rate (the rate at which it lends to commercial banks), which in turn influences both loan and deposit rates in the economy. Thus, we assume that the interest rate is exogenous and money supply is endogenous.

The main variable that influences aggregate demand is the real rate of interest, i.e., the excess of nominal interest rate over (expected) inflation. We shall assume that the Central Bank can influence the short term real interest rate by varying the short term nominal interest rate. A question arises: how does the Central Bank decide on the interest rate?

In macroeconomic literature it is well established that the Central Bank fixes the (real) rate of interest according to the following formula:

$$r = r' + a(x - x') + b(y - y') \quad \text{... (17.11)}$$

The formula given at equation (17.11) is called the Taylor's rule. Equation (17.11) assumes that both $a$ and $b$ are positive. It implies that the rate of interest is positively related to deviation of output $(y - y')$ and deviation of inflation $(x - x')$ from their target levels. When output and inflation are at their target levels, the rate of interest will be equal to $r'$. If actual inflation is higher than targeted inflation (or, actual output is higher than natural output), the Central Bank will increase the rate of interest. You would have noticed this in the decisions of the Monetary Policy Committee of the RBI.

The demand side of the economy is represented by an IS curve of the following type:

$$y - y' = c - dr \quad \text{... (17.12)}$$

Equation (17.12) indicates that actual output and, thus, the output gap responds negatively to real interest rate $(r)$, where $c$ captures the effect of other factors such as fiscal policy and external demand on output. Notice that the output gap will be zero when $r = \frac{c}{d}$. The rate of interest $(r)$, at which output gap equals zero is called as the 'neutral rate of interest'. Both $c$ and d are, of course, positive.

Inflation and output are related to each other through an "accelerationist" Phillip's Curve. Inflation is of the demand-pull variety. Inflation increases over time when high aggregate demand causes output to remain above its natural level (this requires $e$ to be positive in equation (17.13)). This is captured by the following simple expression:

$$\Delta x = e(y - y') \quad \text{... (17.13)}$$

The difference operator (Δ) shows the change in the value of a variable between two time periods. Notice that equation (17.13) can be derived from equation (17.2) by assuming that the public expects the previous year's inflation rate. That is, equation (17.13) can be viewed as an adaptive expectations version of equation (17.2).

Equations (17.11) and (17.12) suggest a positive relationship between inflation rate and output. When inflation rate increases, the Central Bank responds by increasing the real interest rate. This, in turn, contracts aggregate demand and total output in the economy. By using equation (17.12) in equation (17.11), we obtain the following equation of the aggregate demand (AD) curve:

$$y = y' + \left[\frac{c - dr' + dax'}{1+bd}\right] - \frac{dax}{1+bd} \qquad \text{... (17.14)}$$

Aggregate demand is depicted in Fig. 17.2. Macroeconomic equilibrium will be attained at that point on the AD line where output and inflation rate are at their respective targets. Clearly, none of the variables will then have any tendency to change. Since output is pegged at its natural level, inflation rate will remain unchanged due to our formulation of the Phillips Curve in equation (17.12). Also, since output and inflation targets have been achieved, the Central Bank will have no incentive to change the rate of interest. Thus, the combination of $y'$ and $x'$ on the AD curve in Fig. 17.2 represents an equilibrium condition. We observe from equation (17.14) that if inflation is $x'$, output will be equal to $y'$ when the following condition holds hold

$$r' = \frac{c}{d} \qquad \text{... (17.15)}$$

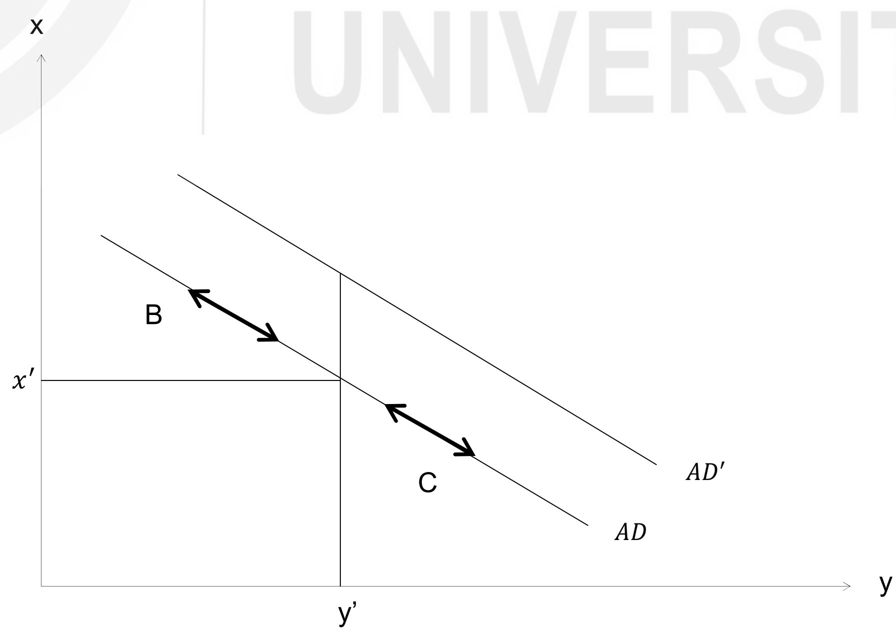


**Fig. 17.2: Macroeconomic Equilibrium**

The Central Bank must set parameter $r'$ equal to the neutral rate of interest. This may not be simple, as the Central Bank may not be aware of the correct parameters of the IS curve and may end up underestimating or overestimating the neutral rate. Since $y = y'$ at the equilibrium point, it follows from equation (17.1) that:

$$r = r' = \frac{c}{d} \qquad \text{... (17.16)}$$

If the Central Bank fixes the value of $r'$ according to equation (17.15), the aggregate demand (AD) curve will pass through $(y', x')$ in Fig. 17.2. If $r' \neq \frac{c}{d}$, then it must be the case that $x \neq x'$ when $y = y'$. This can be verified from equation (17.14). In Fig.17.2, such a situation arises when aggregate demand is given by the line $AD'$.

Now, suppose the Central Bank fixes $r'$ according to equation (17.15) so that aggregate demand is given by the AD line in Fig. 17.1. If the economy lies to the left of point $(y', x')$ on the AD line, actual output will remain below its natural level. Therefore, due to our formulation of the Phillips' Curve, inflation rate must be falling. The Central Bank will then swing into action by reducing the rate of interest and, thereby, expanding aggregate demand. Thus, from B, the economy will move down along the AD line as indicated by the arrow pointed in the south-east direction. At B, actual output is less than the natural output and, thus, it follows from equation (17.12) that actual interest rate must exceed the natural interest rate fixed by equation (17.15). For any point to the right of $(y', x')$ on the AD line, inflation must be rising causing our hypothetical economy to move in the north-west direction. Similarly, the actual interest rate must be less than the neutral interest rate. On the whole, the Central Bank has to first identify the correct neutral rate of interest and then vary the actual rate of interest towards the neutral rate.

Let us now examine the impact of two policy experiments: (i) an increase in inflation target of the Central Bank, and (ii) a shift in fiscal policy.

### 17.3.1 Impact of a Higher Inflation Target

Suppose the economy is initially at equilibrium point A in Fig. 17.3 when the Central Bank increases its inflation target $x'$. This causes the AD line to shift to $AD'$ in Fig. 17.2. It can be verified from equation (17.5) that the intercept of AD line increases due to an increase in $x'$. Intuitively, this happens because a higher inflation target results in a lower rate of interest due to interest rate rule. Note that a shift in inflation target does not affect the neutral rate of interest in the economy, which remains specified according to equation (17.15). The first impact of a higher inflation target is felt by aggregate demand. There is an increase in aggregate demand as actual interest rate falls below its neutral level. This is captured by a movement from A to C in Fig. 17.3. At C, the actual output level exceeds the natural output level causing inflation rate to increase and output to fall along the $AD'$ line. The new equilibrium is attained at B where inflation is at its new target, $x''$. At B, actual output equals natural output and, since the neutral rate of interest remains unchanged, actual inflation is equal to its (new) target (Verify from equation (17.14)). Thus, fixing a higher inflation target results only in a temporary reduction in the interest rate. It causes a short-run increase in aggregate demand and output. In the long run, rate of interest cannot diverge from its neutral level.

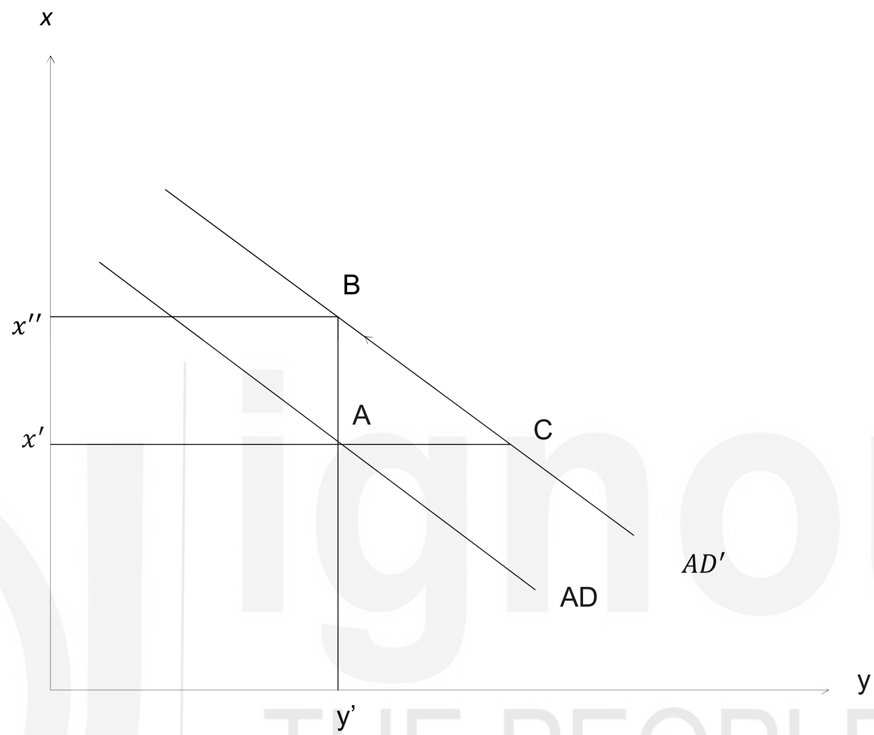


**Fig. 17.3: Impact of a Higher Inflation Target**

### 17.3.2 Impact of Expansionary Fiscal Policy

Let us assume that an increase in government expenditure causes the value of the parameter $c$ to rise. This results in a higher aggregate demand for the given level of inflation and an outward shift of aggregate demand schedule from AD to $AD'$ in Fig. 17.4. Thus, the economy moves from A to C. Inflation rate will now increase causing output to decrease along the $AD'$ line. The economy would thus reach B. However, B in Fig. 17.4 is not an equilibrium point. This is because actual inflation rate now exceeds the target inflation rate. What should the Central Bank do to bring inflation back to its target? Notice that an increase in the value of parameter $c$ increases the neutral rate of interest. As a result, $r'$ must increase to ensure that inflation is at its target when output attains its target. Thus, the Central Bank increases the value of the parameter $r'$, which leads to a contraction of aggregate demand. The economy now moves from B to D in Fig. 17.4. From D, inflation falls and output increases along the AD line till we are back at the original equilibrium point, A.

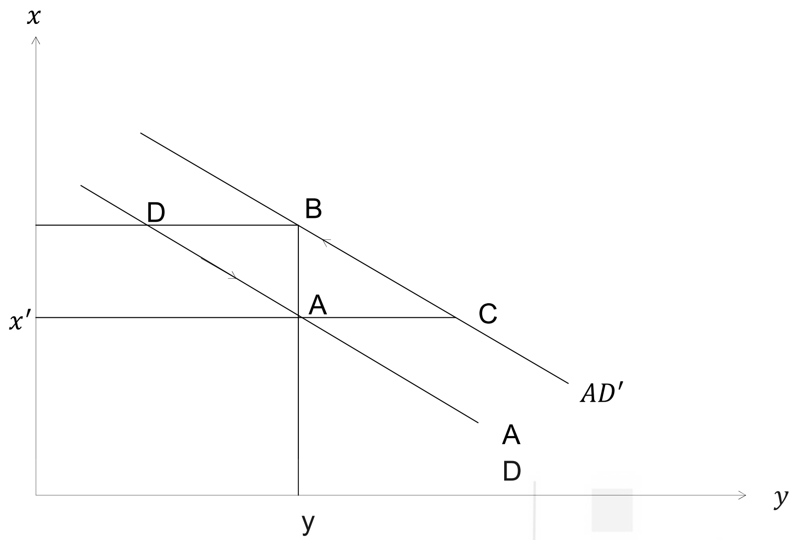


**Fig. 17.4: Impact of Expansionary Fiscal Policy**

An expansionary fiscal policy has no impact on either output or inflation in the economy (Notice that this view is in sharp contrast with the Keynesian view discussed in Unit 2). The only difference that an expansionary fiscal policy makes is to cause a higher interest rate at the same configuration of output and inflation (since the neutral rate of interest, which prevails at the equilibrium, has increased). Thus, expansionary fiscal policy results in a 'crowding out' of private expenditure. In the standard analysis of crowding out, higher public investment causes an excess of total (public and private) investment demand over savings supply. As a result, interest rate increases to equate investment with savings. In the new consensus view, interest rates are fixed by the Central Bank. However, an expansionary fiscal policy results in a higher inflation rate. The Central Bank is thus forced to push up the interest rate to maintain inflation at its target level.

We can sum up the main points of discussion so far. (i) Central Banks can play an important role in stabilising output at its natural level and inflation at some pre-defined target. (ii) Changes in monetary policy aimed at stimulating output above its natural level are bound to fail in the long run. If the Central Bank sets a higher inflation target, it only manages to achieve a higher rate of inflation with no improvement in output. (iii) The Central Bank may also try to reduce interest rate by fixing the rate of interest below the neutral rate of interest. But this policy will also fail to deliver any long run improvement in the output level (Why? What will be the impact of such a policy on inflation rate? Use the diagrams presented above). (iv) There is no space for an activist fiscal policy in the economy. The fiscal policy is also not to be entrusted with the role of controlling inflation and stabilising output levels. These objectives are to be pursued by a Central Bank armed with an appropriate interest rate rule.

**Check Your Progress 2**

1. Describe the consensus view of monetary policy..

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2. Explain, how a higher inflation target influences the interest rate and output? How, these impacts are different from that of an expansionary fiscal policy?

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## 17.4 PATH DEPENDENCE IN MACROECONOMIC OUTCOMES

So far we have assumed that the level of natural output has been exogenously given. Many economists including M. Lavoie have pointed out that natural output level may actually depend on the level of aggregate demand and actual output. First, high actual output promotes learning-by-doing and contributes to improvement in labour productivity or reduction in per unit labour cost. Second, higher output levels induce more capital formation. Higher capital stock in turn improves labour productivity and reduces labour costs. Third, higher aggregate demand may encourage potential workers to enter the labour force and augment labour supply, which must also positively influence the level of natural output. All these factors must push up natural output level of the economy. Simply put, the capacity to produce depends on the amount actually produced. Therefore, we extend the model of the previous section by adding the following equation:

$$\Delta y' = f(y - y') \qquad \text{... (17.17)}$$

Equation (17.17) indicates that the change in the level of natural output is directly or positively related to the output level and hence, the deviations of output. Whenever higher demand pushes actual output above its natural level, potential output increases. Let us assume that the Central Bank always fixes $r'$ according to equation (17.15). Aggregate demand schedule given by equation (17.14) can then be re-written as:

$$y = y' + \left[\frac{da(x'-x)}{1+bd}\right] \qquad \text{... (17.18)}$$

On taking first difference, we obtain:

$$\Delta y = \Delta y' - \left[\frac{da\Delta x)}{1+bd}\right] + \left[\frac{da\Delta x')}{1+bd}\right]$$

We can get the expressions for $\Delta y'$ from equation (17.17) and $\Delta x$ from (17.13) to use them in the above equation. This gives:

$$\Delta y = \left[\frac{f-dae)}{1+bd}\right](y - y') + \left[\frac{da\Delta x')}{1+bd}\right] \qquad \text{... (17.19)}$$

Inflation target $x'$ is constant till the Central Bank changes its policy; $\Delta x'$ is therefore zero for a given policy stance of the Central Bank. Then, it is obvious from equations (17.17) and (17.19) that when $y = y'$ neither $y$ nor $y'$ will have any tendency to change. When the economy reaches the point where $y = y'$, it will tend to stay there. In Fig. 17.5, all points on the $y = y'$ line represent equilibrium combinations of actual output and natural output levels.

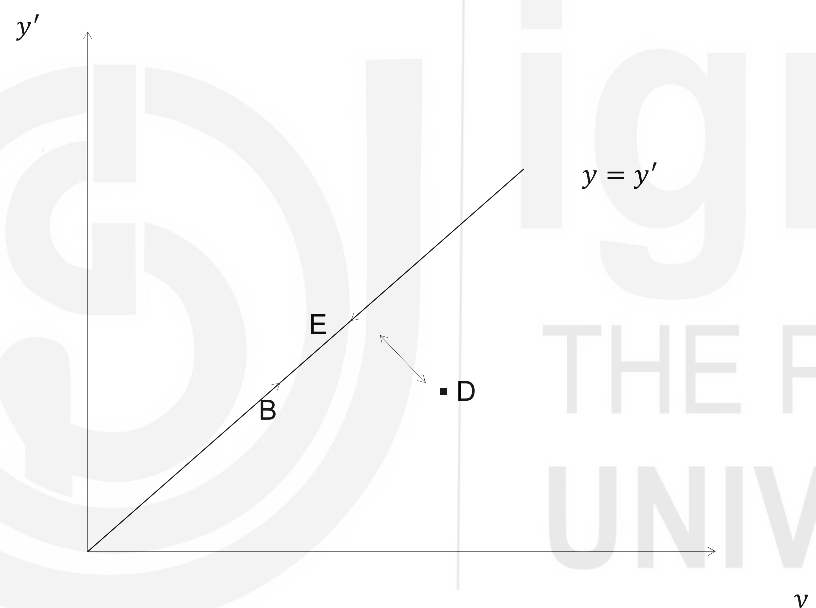


**Fig. 17.5: Path Dependence**

Suppose the economy is initially at B in Fig. 17.5. In the absence of any shock to the system, the economy will stay at B. Now, assume that the Central Bank changes its policy by declaring a higher inflation target. As a result $\Delta x'$ (and, consequently $\Delta y$) would become positive. The economy thus temporarily shifts from B to D in Fig. 17.5. The intuition is that a higher inflation target results in a reduction in interest rate that stimulates aggregate demand and output. As long as we assume that the square bracket term in equation (17.19) is negative, $\Delta y$ would become negative once the economy reaches D. However, the economy will not settle at B again. This is because $y > y'$ at D so that $y'$ will be increasing according to equation (17.17). The economy would thus move in the north-west direction till it reaches the new equilibrium, E. Both $y$ and $y'$ are higher at E compared to B. This model displays path dependence. In other word, there is no unique equilibrium point and the equilibrium we reach depends on the initial out-of-equilibrium point we start from.

We have to be careful about one possibility in the above model. Factors that cause natural output to increase in response to high aggregate demand may take time to kick in. Such factors could be the entry of potential workers in the labour force, immigration of workers from other regions and improvements in labour productivity. If, in the meanwhile, excess demand pressures cause rapid acceleration of the inflation rate so that the Central Bank is also compelled to increase the rate of interest sharply, deviations from natural output level would be short-lived. We need to challenge this possibility because mainstream, following Friedman, believe that natural rate of unemployment has a knife-edge property. Any deviations from it would cause rapid acceleration or deceleration of inflation. In our model, high values of the parameter $e$ (see equation (17.19)) may lead to such an outcome.

There are a number of studies that show that this is not the case. Most studies show that even if unemployment is reduced below its natural level for extended periods of time (natural unemployment corresponds to the natural output level; it is believed to exist because of frictions in the labour market) inflation rate increases only mildly. There is evidence on the shape of the short-run Phillips Curve, which shows some trade-off between unemployment and inflation (in the long run, as you know, there is no trade-off between unemployment and inflation and Phillips Curve is vertical at the natural rate of unemployment). However, the short-run Phillips Curve is usually concave. Thus, unemployment can be driven below its natural rate at a low cost in terms of inflation.

The idea of path dependence (also called hysteresis) in macroeconomic outcomes radically alters the conclusions of the new consensus macro model developed in the previous section. In the new consensus, as in the monetarist and new-classical views, actual output is pegged to an exogenously given natural output. Thus, fiscal and monetary policy interventions designed to increase the level of output are bound to fail. However, once we endogenise the potential or natural output of the economy, activist policies regain their potency. This is the main message of this section.

**Check Your Progress 3**

1. Explain the path dependence of outcomes in macroeconomics.

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## 17.5 LET US SUM UP

In this unit we discussed the rules vs. discretion debate. According to Milton

Friedman, Central Banks should follow a policy rule of constant growth in money supply. Other economists such as Robert Barro defined the rules as a commitment to zero/low inflation policy. For most of the new-Keynesians, the rules are shown in terms of equations that link a Central Bank's 'policy rate' to deviations from inflation and output targets. We observe that that much of the mainstream in economics prefers rules, however understood, to a discretionary use of policy by Central Banks.

Subsequently, we discussed the current consensus on monetary policy. In the consensus model, monetary policy plays a major role in fixing inflation rate at its pre-defined target. In other words, monetary policy can successfully provide the nominal anchor. Moreover, monetary policy, while successful in determining the rate of inflation, cannot influence the level of output, which is fixed at its natural level. In this sense, the new consensus is simply a rehash of the Friedmanite idea that market forces by themselves deliver full employment and monetary policy is neutral. At best, Central Banks can hope to reduce fluctuations of output around its natural level by following an appropriate interest rate policy.

Finally, we discussed the criticism of the new consensus model as it assumes that natural output level is fixed. Critics point out that natural output level may not be exogenously given. High aggregate demand (injected through monetary or fiscal means) may increase actual output as well as natural output levels and take the economy to a higher equilibrium.

## 17.6 ANSWR/HINTS TO CHECK YOUR PROGRESS EXERCISE

**Check Your Progress 1**

1. Refer to Section 8.2. Explain by using equations (17.1) to (17.10) and Fig. 17.1.
2. Refer to Fig. 17.1. Describe why the government should not change its policy stance afterwards.

**Check Your Progress 2**

1. The rate of interest is argued as the main instrument of the Central Bank to control inflation and output. You should explain the Taylor's rule, equations from (17.11) to (17.16) and Fig. 17.2. Refer to Section 8.3.
2. Refer to Section 8.3. Explain by using Fig. 17.3 and Fig. 17.4.

**Check Your Progress 3**

1. The natural output level may not be exogenously given. It may actually depend on the level of aggregate demand and actual output. First, high actual output promotes learning-by-doing and contributes to improvement in labour productivity or reduction in per unit labour cost. Also, higher output levels induce more capital formation. Higher capital stock in turn improves labour productivity and reduces labour costs. All these factors must push up natural output level of the economy. At the same time the high demand may encourage potential workers to enter the labour force and augment labour supply, which must also positively influence the level of natural output. Explain equations (17.17), (17.18) and (17.19), and Fig. 17.5.